BME364 Group I in the Davos laboratory, 10 to 13 November 2026

Green is TEER, purple is FITC-dextran flux, orange is MTT and blue is ZO-1 immunofluorescence. Yellow is a demonstration, grey is a break, and white is an orientation or a step shared by several assays.

Tuesday 10 November

Time	What we do	Manual and section	Where
16.45–16.55	General laboratory safety introduction (Hang Du)		Meeting room 2nd floor
16.55–17.25	SIAF laboratory tour in three groups		Laboratories

Wednesday 11 November

Time	What we do	Manual and section	Where
08.15–09.00	Overview of the experiments of all three groups		Meeting room 2nd floor
09.00–10.30	• Harvest the Caco-2 cells and count them • Seed the cells in the 96-well plate	MTT manual, Harvest Cells to Cell seeding	Lab
10.30–10.45	Coffee break		
10.45–11.15	• Prepare the P80 stock • Prepare the P80 treatment solutions • Sterilise the TEER electrode	TEER manual, Solutions and Electrode Preparation IF manual, Solutions	Lab
11.15–12.00	Measure TEER at 0 h	TEER manual, Measuring One Time Point and Baseline and Treatment (0 h)	Lab
12.00–12.30	Add P80 to the Transwell inserts	TEER manual, Baseline and Treatment (0 h)	Lab
12.30–13.30	Lunch break		
13.30–14.00	Overview of the IF experiment	IF manual	Meeting room 2nd floor
14.00–15.00	Demonstration of nasal organoids		Lab
15.00–15.30	Add P80 to the IF glass-bottom plate	IF manual, Cell Seeding and Treatment	Lab
15.30–15.45	Coffee break		
15.45–16.45	• Demonstration of the gut-on-a-chip model • TEER measurement on the gut-on-a-chip		Lab

Thursday 12 November

Time	What we do	Manual and section	Where
09.30–10.15	• Prepare the P80 dilution series • Prepare the vehicle control	MTT manual, Solutions (Serial Dilution of P80)	Lab
10.15–10.45	Treat the cells with P80	MTT manual, Treatment	Lab
10.45–11.00	Coffee break		
11.00–11.20	Prepare the FD4 dosing solutions	Flux manual, Solutions	Lab
11.30–12.15	Measure TEER at 24 h	TEER manual, 24 h and 48 h	Lab
12.15–12.45	Replace the medium and add FD4	Flux manual, FD4 Addition	Lab
12.45–13.45	Lunch break		
13.45–14.45	Demonstration of the skin barrier measured by electrical impedance spectroscopy (EIS)		Lab
14.45–15.00	Coffee break		
15.00–15.30	Fix the cells and wash them	IF manual, Fixation	Lab
15.30–15.50	• Permeabilise the cells and wash them • Start the blocking	IF manual, Permeabilisation and Blocking	Lab
16.50–17.05	Add the primary antibody and leave it overnight	IF manual, Antibody and DAPI Staining	Lab

Friday 13 November

Time	What we do	Manual and section	Where
08.45–09.15	• Wash off the primary antibody • Add the secondary antibody	IF manual, Antibody and DAPI Staining	Lab
09.15–10.00	Prepare the FD4 standard curve	Flux manual, Standard Curve	Lab
10.00–10.15	Add the MTT reagent	MTT manual, Adding MTT reagent	Lab
10.15–10.45	• Wash the cells • Stain the nuclei with DAPI	IF manual, Antibody and DAPI Staining	Lab
10.45–11.00	Coffee break		
11.00–11.30	Introduction to the confocal microscope	IF manual, Confocal Imaging	Confocal room
11.30–12.15	• Measure TEER at 48 h • Clean the electrode	TEER manual, 24 h and 48 h	Lab
12.15–12.30	Take the apical and basolateral FD4 samples	Flux manual, Sampling	Lab
12.30–13.30	Lunch break (group pizza lunch)		
13.30–13.45	Read the FD4 plate	Flux manual, Sampling	Lab
13.45–14.15	• Dissolve the formazan • Read the MTT plate	MTT manual, Measurement	Lab
14.15–16.30	Image the IF plate on the confocal microscope	IF manual, Confocal Imaging	Confocal room
16.30–17.00	Clean up the laboratory		Lab